

RM 510: Research Methodology

Course Code	RM 510
Course Name	Research Methodology
L-T-P-C	1-0-0-1
Prerequisites	None
Intended for	MS/PhD
Mutual Exclusion	None
Approval	9th Senate (May 2016)

Course Contents

Each topic below represents one lecture.

- **Introduction:** What is research, how research has benefited human society
- **Literature survey:** What is it? Why is it needed? How to do a literature survey, how to read a research paper, use of tools like Google Scholar, CiteSeer etc.
- **Ph.D degree and thesis:** What are the objectives, and what are not. What is the role of the advisor and the scholar?
- **Things that may help a research scholar:** for example, maintain a daily routine, attend talks in other areas, go for others' Ph.D. seminars, maintain physical fitness, time management etc.
- **Research ethics:** plagiarism, academic dishonesty, a few case studies
- **Writing:** How to write good papers, and research proposals
- **Presentation:** How to make good presentations
- **School-specific lecture 1**
 - SHSS: Theory (broad topics, specific to the discipline of the student)
 - SBS: Important sources of literature -journals and conferences of RSC, ACS, AMS, IOP, APS, ScienceDirect, use of different reference tools
 - SCEE: Important sources of literature- journals and conferences of ACM, IEEE etc. • SE: Intellectual property rights
- **School-specific lecture 2**
 - SHSS: Practice (broad topics, specific to the discipline of the student)
 - SBS: Introduction to software like ChemDraw, SolidWorks- 1
 - SCEE: Using computing tools: HPC cluster, LaTeX, Matlab, producing figures and plots
 - SE: Good practices for numerical simulations
- **School-specific lecture 3**
 - SHSS: Writing (broad topics, specific to the discipline of the student)
 - SBS: Introduction to software like ChemDraw, SolidWorks-2
 - SCEE: Reproducible research in computer science and electrical engineering
 - SE: Good practices for experimental methods
- Writing a research proposal and making presentation. Each student will prepare a 3-page proposal and give a 10-minute presentation on it.
- Exercise 1
- Exercise 2
- Exercise 3
- Exercise 4

Note on abbreviations

- SHSS: School of humanities and social sciences
- SBS: School of basic sciences
- SCEE: School of computing and electrical engineering
- SE: School of engineering

Text Books

References

1. Richard Hamming, **You and your research**, Bell Communications Research Colloquium Seminar, 1986 (Transcription available online.)
2. Cone, J.D. & Foster, S.L., **Dissertations and theses from start to finish**, DC, APA, 2006.
3. Galvin, J.L., **Writing literature reviews: A guide for students of the social and behavioral sciences**, 4th Edition, Pyczak, 2009.
4. Gilpin, A.A., **A guide to writing in the sciences**, U. Toronto Press, 2000.
5. Gustavii, B., **How to write and illustrate a scientific paper**, Cambridge University Press, 2003.
6. Babbie, E.R., **The practice of social research**, Wadsworth, 2010.